PROJECT REPORT

On

"Deep Learning for Lung Disease Detection from CT Images"

Submitted in partial fulfilment of the requirements for the award of

Bachelor of Technology (B. TECH)

In the department of

Computer Science and Engineering

Submitted by:

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CERTIFICATE

This is to certify that the project report entitled *"*Deep Learning for Lung Disease Detection from CT Images*"*, submitted to the School of Engineering & Technology (SOET), **ADAMAS UNIVERSITY, KOLKATA** in partial fulfilment for the completion of **Semester – 6th** of the degree of **Bachelor of Technology** in the department of **Computer Science and Engineering**, is a record of bonafide work carried out by **Shubhabrata Choudhuri (UG/02/BTCSE/2023/140), Parthib Nandi (UG/02/BTCSEAIML/2023/010), Suvranil Ghatak(UG/02/BTCSEAIML/2023/015)** under our guidance.

All help received by us from various sources have been duly acknowledged.
No part of this report has been submitted elsewhere for award of any other degree.

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Finally, we express our gratitude to all other members who are involved either directly or indirectly for the completion of this project.

DECLARATION

We, the undersigned, declare that the project entitled "Deep Learning for Lung Disease Detection from CT Images", being submitted in partial fulfillment for the award of Bachelor of Technology Degree in Computer Science and Engineering, affiliated to ADAMAS University, is the work carried out by us.

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ABSTRACT

Lung cancer is one of the leading causes of death worldwide, and early detection is essential for effective treatment and improved patient survival. Manual analysis of CT scan images by radiologists is time-consuming and may sometimes lead to diagnostic errors. Therefore, there is a growing need for automated systems that can assist in accurate and efficient lung cancer detection.

This project, titled "Deep Learning Based Lung Cancer Detection Using CT Scan Images", focuses on developing an AI-based medical image classification system using deep learning techniques. The IQ-OTHNCCD Lung Cancer Dataset was used in this project, which contains CT scan images categorized into Normal, Benign, and Malignant cases.

The proposed system uses Convolutional Neural Networks (CNNs) to automatically extract important features from CT scan images and classify them into Normal and Cancer categories. Various preprocessing techniques such as grayscale conversion, image resizing, normalization, and data augmentation were applied to improve model performance and reduce overfitting.

The CNN model was trained and evaluated using performance metrics such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix. Experimental results demonstrate that deep learning techniques can effectively assist in automated lung cancer detection and medical image analysis.
This project highlights the practical application of Artificial Intelligence in healthcare and demonstrates the potential of deep learning systems for intelligent medical diagnosis support.

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INTRODUCTION

1.1 Background:

Lung cancer is one of the most dangerous diseases affecting people worldwide. Early detection of lung cancer is very important because it helps doctors provide proper treatment and improves patient survival rates. CT (Computed Tomography) scan imaging is widely used for detecting abnormalities inside the lungs because it provides detailed internal images.

Traditional diagnosis of CT scan images is performed manually by radiologists, which is time-consuming and may sometimes lead to human errors. With the rapid growth of Artificial Intelligence (AI) and Deep Learning, automated medical image analysis systems have become highly effective in disease detection.

Convolutional Neural Networks (CNNs), a popular deep learning technique, have shown excellent performance in image classification tasks. In this project, a CNN-based deep learning model is developed to detect lung cancer from CT scan images using the IQ-OTHNCCD Lung Cancer Dataset.

1.2 Purpose of the Project:

The main purpose of this project is to develop an automated lung cancer detection system using CT scan images and deep learning techniques.

The system aims to:

- Assist radiologists in diagnosis
- Improve detection accuracy
- Detect lung cancer at an early stage
- Reduce manual workload
- Demonstrate AI applications in healthcare

1.3 Problem Statement:

Manual analysis of CT scan images requires expert knowledge and consumes a significant amount of time. Large numbers of medical scans in hospitals increase the workload of radiologists and may cause delays in diagnosis.

Some major challenges of manual diagnosis include:

- Human errors in interpretation
- Slow diagnosis process
- High workload for medical experts
- Limited accessibility to healthcare support

Therefore, an automated deep learning-based system is required for efficient and accurate lung cancer detection using CT scan images.

1.4 Objective:

The main objectives of this project are:

- Collect CT scan images from the IQ-OTHNCCD Lung Cancer Dataset
- Preprocess CT scan images for deep learning
- Develop a CNN-based lung cancer detection model
- Classify CT images into Normal and Cancer categories
- Evaluate model performance using Accuracy, Precision, Recall, F1-score, and Confusion Matrix
- Develop an intelligent AI-based healthcare support system

1.5 Structure of the Project:

This project report is divided into multiple chapters. Chapter 1 introduces the project background, objectives, and problem statement. Chapter 2 presents the literature review related to deep learning and lung cancer detection. Chapter 3 explains the technologies and tools used in the project. Chapter 4 discusses the results and performance analysis of the CNN model. Chapter 5 describes the methodology used for implementation. Chapter 6 presents the conclusion of the project, while Chapter 7 discusses future improvements and research scope.

Chapter 2
LITERATURE REVIEW

Recent advancements in Artificial Intelligence (AI) and Deep Learning have significantly improved medical image analysis, especially in lung disease detection using CT scan images. Researchers worldwide are developing automated systems that can identify lung abnormalities accurately and efficiently.

Convolutional Neural Networks (CNNs) are widely used in medical imaging because they can automatically extract important features from CT scan images and classify normal and abnormal lung conditions. Many studies have shown that CNN-based models achieve high accuracy in detecting lung cancer and pulmonary nodules.

Several researchers have also used transfer learning models such as VGG16, ResNet, and DenseNet to improve classification performance and reduce training time. These pre-trained models help improve feature extraction and increase prediction accuracy.

Some studies focused on 3D CNN architectures for analysing volumetric CT scans. Compared to traditional 2D CNNs, 3D CNNs can capture more spatial information from multiple CT slices. However, these models require high computational power and larger datasets.

Researchers have also introduced attention mechanisms and Explainable AI (XAI) techniques such as Grad-CAM to improve model interpretability and highlight important regions in CT scan images responsible for predictions.

Despite significant improvements, some limitations still exist in AI-based lung disease detection systems. Challenges such as limited datasets, overfitting, high computational requirements, and low explainability affect model performance and real-world implementation.

Overall, deep learning techniques, especially CNN-based models, have shown strong potential in automated lung cancer detection using CT scan images. Future research can further improve accuracy, efficiency, and interpretability of these systems for real-world healthcare applications.

Chapter-3
TECHNOLOGY USE

3.1 Overview:
The proposed lung disease detection system is developed using Deep Learning and Computer Vision technologies. The system uses CT scan images and Convolutional Neural Networks (CNNs) to classify lung conditions into normal and cancerous categories. Various Python libraries and frameworks were used for image preprocessing, model development, training, and evaluation.

3.2 Programming Language:
Python was used as the primary programming language for developing the project because of its simplicity, flexibility, and large collection of machine learning libraries.

3.3 TensorFlow and Keras:
TensorFlow and Keras were used for designing and training the Convolutional Neural Network (CNN) model.

The CNN model consists of:

- Convolution layers
- Pooling layers
- Batch normalization
- Dropout layers
- Dense layers

These layers help extract features from CT scan images and perform classification.

3.4 OpenCV:
OpenCV was used for image preprocessing operations such as:

- Reading CT images
- Converting images into grayscale
- Resizing images
- Image normalization

OpenCV improves image quality and prepares images for CNN training.

3.5 NumPy:
NumPy was used for numerical computations and array manipulation during image processing and dataset preparation.

3.6 Matplotlib and Seaborn:
Matplotlib and Seaborn libraries were used for:

- Accuracy graph plotting
- Loss graph plotting
- Confusion matrix visualization

These visualizations help analyze model performance.

3.7 Scikit-Learn:
Scikit-Learn was used for:

- Train-test splitting
- Classification report generation
- Confusion matrix calculation
- Performance evaluation metrics

3.8 Google Colab:
Google Colab was used as the cloud-based development environment for training and testing the deep learning model. GPU support provided faster computation and model training.

3.9 Dataset:
The IQ-OTHNCCD Lung Cancer Dataset was used in this project. The dataset contains CT scan images divided into:
Normal cases
Benign cases
Malignant cases
The images were used for binary classification between normal and cancer conditions.

Chapter-4

Result and Analysis

4.1 Overview:

This chapter presents the results obtained from the proposed deep learning-based lung disease detection system. The CNN model was trained and evaluated using CT scan images from the IQ-OTHNCCD dataset.

4.2 Dataset Summary:

The dataset contains CT scan images categorized into:

- Normal cases
- Benign cases
- Malignant cases

For binary classification:

- Normal images were labelled as 0
- Benign and Malignant images were labelled as 1 (Cancer)

Total images used: 1097

Image size after preprocessing: 128 × 128 pixels

4.3 Image Preprocessing Results:

The following preprocessing techniques were applied:

- Image resizing
- Grayscale conversion
- Normalization
- Data augmentation

These preprocessing steps improved training efficiency and reduced overfitting.

4.4 CNN Model Architecture:

The CNN model consists of:

- 5 Convolutional layers
- MaxPooling layers
- Batch normalization layers
- Dropout layers
- Fully connected Dense layers

The final output layer uses Sigmoid activation for binary classification.

4.5 Training Results:

The model was trained using:

- Optimizer: Adam
- Loss Function: Binary Cross entropy
- Epochs: 10
- Batch Size: 32

The training process achieved stable learning and reduced loss over epochs.

4.6 Performance Evaluation:

The trained model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Final Test Accuracy:

- Approximately 76%

The model showed good performance in detecting cancerous CT images.

4.7 Confusion Matrix Analysis:

The confusion matrix shows:

- Correctly classified normal cases
- Correctly classified cancer cases
- Misclassified samples

The model successfully identified most cancer cases while maintaining acceptable normal classification performance.

4.8 Accuracy and Loss Graphs:

Training and validation accuracy graphs show the learning behavior of the CNN model.

Loss graphs demonstrate how the model minimized prediction error during training.

4.9 Advantages of the System:

- Automated lung disease detection
- Faster diagnosis support
- Reduced manual workload
- Efficient CT image analysis
- AI-based medical assistance

4.10 Limitations:

The system has certain limitations:

- Small dataset size
- Possibility of overfitting
- Limited generalization
- Requires high computational resources

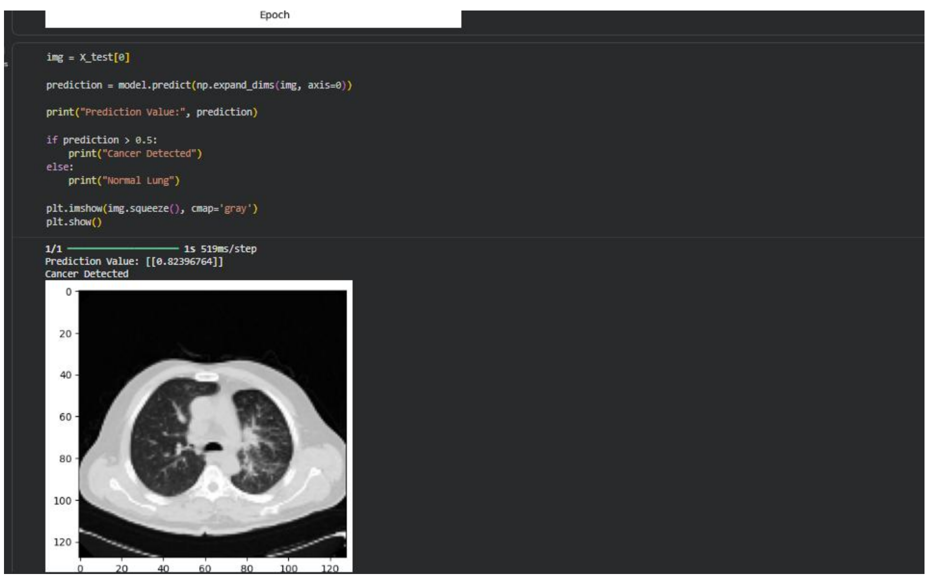


Fig1: Serial Monitor Output

```
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])

plt.title('Model Loss')
plt.ylabel('Loss')
plt.xlabel('Epoch')

plt.legend(['Train', 'Validation'])

plt.show()
```

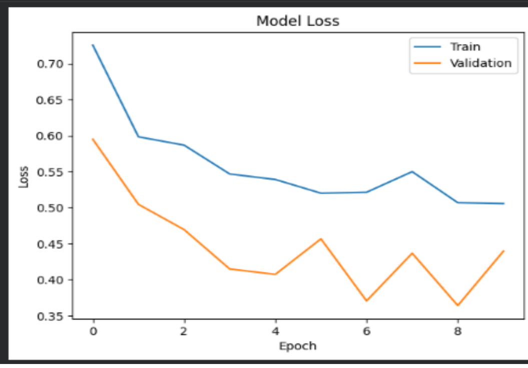


Fig2: Model loss Graph

```
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])

plt.title('Model Accuracy')
plt.ylabel('Accuracy')
plt.xlabel('Epoch')

plt.legend(['Train', 'Validation'])

plt.show()
```

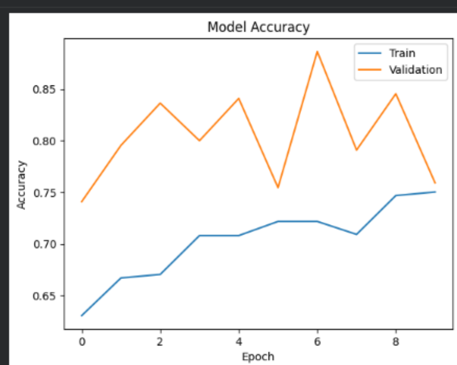
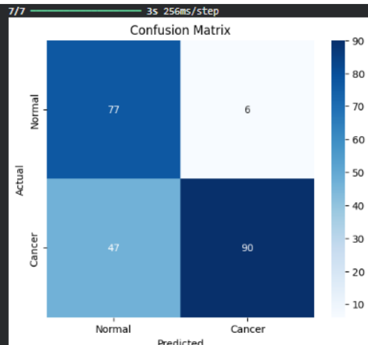


Fig3: Model Accuracy

```
7/7 ━━━━━━━━━━━━━━━━━━━━ 3s 256ms/step
```




```
              precision    recall  f1-score   support

           0       0.62      0.93      0.74        83
           1       0.94      0.66      0.77       137

    accuracy                           0.76       220
   macro avg       0.78      0.79      0.76       220
weighted avg       0.82      0.76      0.76       220
```

Fig4: Confusion Matrix

Chapter-5
METHODOLOGY

The methodology of this project follows a systematic approach for developing a deep learning-based lung cancer detection system using CT scan images. The complete workflow includes dataset collection, image preprocessing, CNN model development, model training, testing, and performance evaluation.

In the first stage, the IQ-OTHNCCD Lung Cancer Dataset was collected and used for the project. The dataset contains CT scan images categorized into three classes:

- Normal cases
- Benign cases
- Malignant cases

For this project, binary classification was performed. The Normal class was labelled as 0, while both Benign and Malignant classes were labelled as 1 (Cancer).

After collecting the dataset, preprocessing techniques were applied to prepare the CT images for deep learning. The images were loaded using OpenCV in grayscale format. Each image was resized to a fixed dimension of 128 × 128 pixels to maintain consistency throughout the dataset. Image normalization was also performed by dividing pixel values by 255 to scale the values between 0 and 1, which improves model training performance.

The prepared dataset was converted into NumPy arrays and reshaped into a four-dimensional format suitable for Convolutional Neural Networks (CNNs). The dataset was then divided into training and testing datasets using the train-test split technique. Approximately 80% of the data was used for training, while 20% was used for testing and validation.

To improve the generalization capability of the model and reduce overfitting, data augmentation techniques were applied using TensorFlow's ImageDataGenerator. Augmentation operations such as rotation, zooming, horizontal flipping, and shifting were performed on the training images.

In the next stage, a Convolutional Neural Network (CNN) model was developed using TensorFlow and Keras. The CNN architecture consists of multiple convolutional layers, max-pooling layers, batch normalization layers, dropout layers, and fully connected dense layers. The convolutional layers extract important features from CT scan images, while pooling layers reduce spatial dimensions and computational complexity. Batch normalization improves training stability, and dropout layers help prevent overfitting.

The final output layer uses a sigmoid activation function for binary classification between normal and cancerous lung conditions. The model was compiled using the Adam optimizer and Binary Cross entropy loss function.

The CNN model was trained using the augmented training dataset with a batch size of 32 and multiple epochs. During training, the model learned important patterns and features associated with lung abnormalities from CT scan images.

After training, the model was evaluated using the testing dataset. Various performance metrics such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix were used to analyse the effectiveness of the model. The confusion matrix helped visualize correct and incorrect classifications for both normal and cancer classes.

Finally, the trained deep learning model was used to predict lung conditions from unseen CT scan images. The proposed methodology demonstrates the practical implementation of deep learning techniques for automated lung cancer detection and medical image classification.

Chapter-6
CONCLUSION

The project “Deep Learning Based Lung Cancer Detection Using CT Scan Images” successfully demonstrates the use of Deep Learning and Artificial Intelligence for automated medical image classification. In this project, CT scan images from the IQ-OTHNCCD Lung Cancer Dataset were used to classify lungs into Normal and Cancer categories using a Convolutional Neural Network (CNN).

Various preprocessing techniques such as grayscale conversion, resizing, normalization, and data augmentation were applied to improve model performance. The CNN model was trained and evaluated using TensorFlow and Keras, achieving good classification accuracy on the testing dataset.

The system helps in faster and more efficient lung cancer detection compared to manual analysis. Performance evaluation using Accuracy, Precision, Recall, F1-score, and Confusion Matrix shows that deep learning techniques are effective for medical image analysis.

Overall, the project demonstrates the potential of AI-based healthcare systems for early lung cancer detection and diagnostic support. Future improvements can be made using larger datasets and advanced deep learning models to achieve better accuracy and real-world performance.

Chapter-7

FUTURE WORK

The proposed deep learning-based lung cancer detection system can be further improved in several ways to achieve better accuracy, efficiency, and real-world usability.

Future improvements may include:

- Using larger and more diverse CT scan datasets to improve model accuracy and generalization.

- Implementing multi-class classification to detect different lung diseases such as pneumonia, tuberculosis, and COVID-19 along with lung cancer.

- Applying advanced deep learning models such as ResNet, DenseNet, 3D CNN, and Vision Transformers for better feature extraction and classification performance.

- Using Explainable AI (XAI) techniques such as Grad-CAM to visualize important regions in CT scan images responsible for predictions.

- Developing lightweight models for faster execution on low-resource systems and rural healthcare centres.

- Integrating the system into web or mobile healthcare applications for real-time diagnosis and remote medical support.

- Improving model performance using transfer learning and ensemble learning techniques.

Overall, future advancements can make the system more accurate, reliable, and suitable for real-world medical applications and intelligent healthcare systems.

Chapter-8

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